Total No. of Printed Pages: 2

**SUBJECT CODE NO: - RNP-8005**
**FACULTY OF SCIENCE AND TECHNOLOGY**
**S.E (EEE / EE / EEP) (NEP) Sem - III**
**Examination May/June-2026**
**Transformer and DC Machine**

**[Time: 2:00 Hours]** **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1. Part A: Question No 1 is compulsory.
2. Attempt any four questions from remaining Part B: Q.2 to Q.7.

## PART A

**Q1 Solve any 10 following questions** **20**

a) In DC series motor, speed is.............
 a) Almost constant b) Low at no-load
 c) Very high at no-load d) Zero at no-load

b) A DC machine works on the principle of.........................
 a) Lenz's law
 b) Faraday's law of electromagnetic induction
 c) Coulomb's law
 d) Gauss law

c) The function of tap changer is to ...............
 a) Increase frequency b) Control voltage
 c) Reduce power factor d) Increase current

d) In an ideal transformer, input power is equal to........................
 a) Zero b) Output power
 c) Copper loss d) Core loss

e) Efficiency of a transformer is maximum when .......................
 a) Core loss = Copper loss b) Core loss > Copper loss
 c) Copper loss = 0-meter d) Total loss is minimum

f) The core of a transformer is laminated to reduce ................
 a) Hysteresis loss b) Eddy current loss
 c) Copper loss d) Stray losses

g) The transformer works on the principle of .......................
 a) Ohm's law b) Mutual induction
 c) Electrostatic inductions d) Joule's law

h) The starting torque of DC series motor is ....................
 a) Low b) Medium
 c) High d) Zero

i) Which DC generator has the poorest voltage regulation?

a) Shunt generator b) Series generator
c) Compound generator d) Differential compound generator

j) The main function of a yoke in a DC machine is..........................
a) Provide mechanical support b) Complete the magnetic path
c) Protect the machine d) All of the above

k) The EMF generated in a DC generator is:
a) AC b) DC
c) Both AC and DC d) None

l) High starting current in DC motor is due to:
a) High back EMF b) Low armature resistance
c) High field flux d) Mechanical load

## PART B

**Solve any four from the following**

| | | | |
|---|---|---|---|
| **Q.2** | a) | Explain Construction & Working principle of D.C generator | **5** |
| | b) | Explain Construction & Working principle of 1Phase Transformer | **5** |
| **Q.3** | a) | Speed control of DC Shunt motor | **5** |
| | b) | With the help of neat circuit diagram explain how O.C. & S.C. test is conducted on single-phase transformer. | **5** |
| **Q.4** | a) | Why is core of transformer laminated? Explain? | **5** |
| | b) | What are the different types of losses in transformer & explain it? | **5** |
| **Q.5** | a) | Write a Short Note on Three-point starter | **5** |
| | b) | Write a Short Note on Stepper Motor | **5** |
| **Q.6** | a) | A 4 pole DC Shunt generator with lap connected armature has field and armature resistance of 80 $ohm$ and 0.1 $ohm$ respectively. It supplies power to 50 lamps, rated to 100 volts, 60 watt each. Calculate total armature current and generated emf, by allowing a brush drop of $2V$ per brush. | **5** |
| | b) | Explain: Power Flow Diagram of DC Machine | **5** |
| **Q.7** | a) | Explain: Losses in DC Machines. | **5** |
| | b) | Explain: Armature Reaction in DC Machines. | **5** |